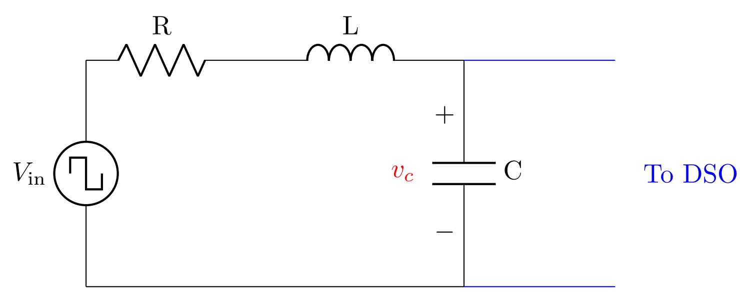
Experiment 5
Step response of second order circuits

Components/Instruments/Meters

- Digital storage oscilloscope (DSO)
- Function generator
- Breadboard
- Resistor
- Capacitor
- Inductor
- Connecting wires

Step response of RLC circuit

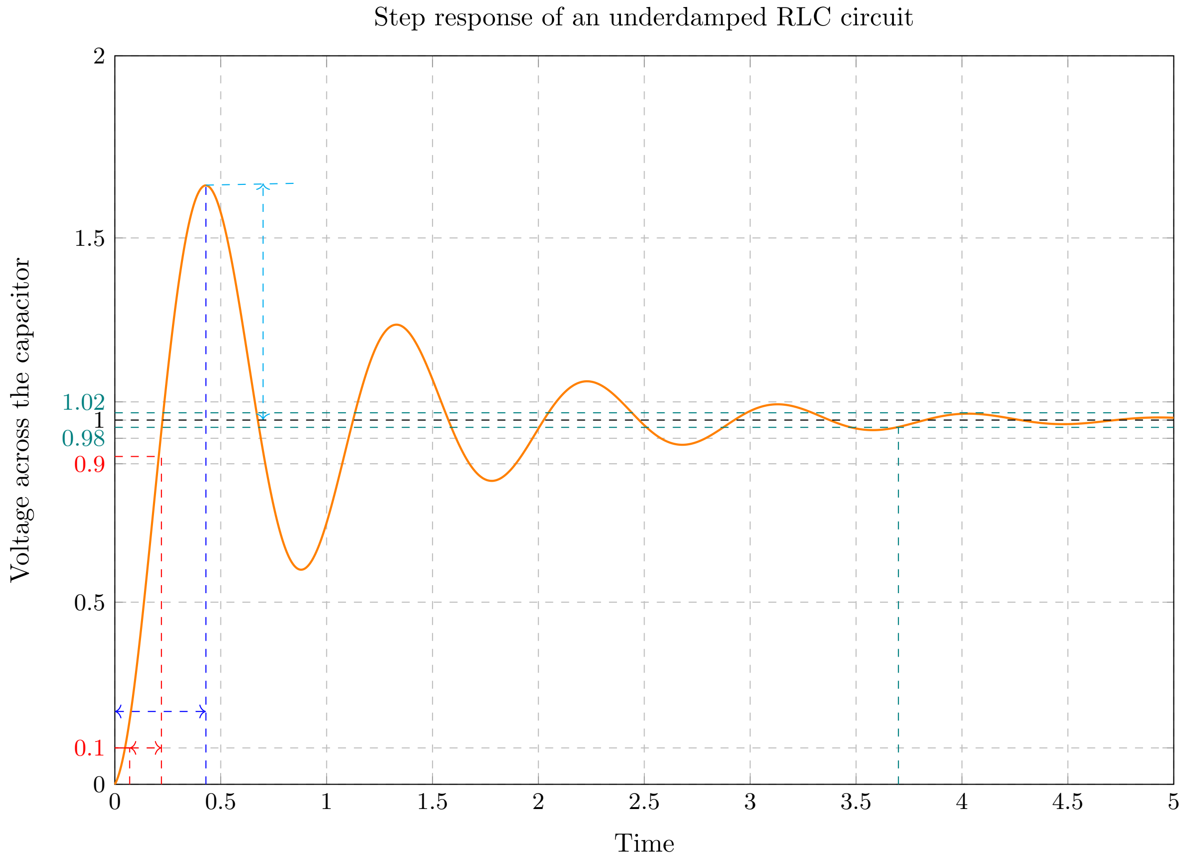
1. Setup a circuit as shown below. Set the value of R as 1 kΩ.



R	C	L
0–5 kΩ	220 nF	2H

2. Set the signal generator to square wave signal (Peak-to-peak value = 2 V, frequency = 50 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit (First terminal of "R" and second terminal of "C").
4. Connect 'Channel-2' of DSO to output terminals of the circuit (First terminal of "C" and second terminal of "C"), i.e., voltage across capacitor "C" which is v_c.

5. From the captured waveform, measure the parameters listed in the table.



Parameters		Values
Rise time (Time to go from 10% to 90%)	$t_r = \dfrac{\pi - \tan^{-1}\dfrac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_n\sqrt{1-\zeta^2}}$	
Peak time (Time of the first peak)	$t_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$	
Settling time (Time to settle within 2% band)	$t_s = \dfrac{4}{\zeta\omega_n}$	
Peak overshoot (Overshoot from steady state)	$M_p = e^{-\dfrac{\pi\zeta}{\sqrt{1-\zeta^2}}}$	

6. Compute the natural frequency (ω_n) and damping ratio (ζ) from your observations.

7. Can you change the value of R to observe an overdamped response on the DSO?

8. Identify the critical value of R at which the system switches from underdamped to an overdamped one. Compare it with the theoretical value.